

MOHAMMAD AQUIL KHAN

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🌐 LinkedIn | 🐙 GitHub | </> LeetCode

Profile

B.Tech Computer Science student with a focus on Machine Learning and Deep Learning. Strong foundation in Data Structures and Algorithms. Familiar with data preprocessing, model building, and developing AI-based applications for real-world problems.

Education

Jaypee University of Engineering and Technology

B.Tech in Computer Science and Engineering (CGPA: 8.04)

2023 – 2027

Guna, India

Creative Convent School

XII (PCM) - MP Board (81%)

2021 – 2022

Tikamgarh, India

Experience

Machine Learning Intern

NASA Harvest, University of Maryland (Remote)

June 2025 – July 2025

Tools: Python, Machine Learning

- Worked on a **machine learning-based crop forecasting project** and gained practical exposure to the use of AI techniques in agriculture and remote sensing applications.
- Assisted in data preparation and analysis using open-source tools for large-scale datasets.
- Collaborated with researchers during the internship and attended regular progress meetings.

Projects

SwachhCity — AI Waste Management Platform | *Deep Learning, ML, MERN* 🔄 **February 2026 – March 2026**

- Developed a React-based web application with a Python backend using YOLOv8 for automated image classification of waste.
- Applied Random Forest algorithms on historical data to predict garbage hotspots for efficient waste management.
- Designed an AI-based filtering system and routing algorithm to optimize path planning for sanitation workers.

AI-Driven Plant Disease Detection | *Python, Deep Learning* 🔄

December 2025 – January 2026

- Developed a CNN model using TensorFlow on the PlantVillage dataset, achieving 98%+ accuracy in detecting multiple plant diseases.
- Built a FastAPI and React-based platform for real-time plant disease detection and monitoring.
- Optimized image processing pipeline to improve model inference speed and prediction accuracy.

AI-Based Resume Screening & Job Matching System | *React.js, Node.js, OpenAI API, Basic NLP* 🔄 **April 2026**

- Developed an AI-powered recruitment platform using NLP and ML for resume parsing, job matching, and candidate ranking.
- Implemented TF-IDF and cosine similarity to match candidate profiles with job requirements based on skills and experience.
- Designed a recruiter-side filtering and ranking system to shortlist applicants efficiently using AI-generated match scores.

Skills

Programming Languages: Python, C++, JavaScript, SQL

Machine Learning: Supervised Learning, Unsupervised Learning, Scikit-learn, Pandas, NumPy

Deep Learning: TensorFlow, Neural Networks, CNN, Basic NLP

Databases: MySQL

Tools & Platforms: Git, GitHub, Postman, Visual Studio Code, Jupyter Notebook

Relevant Coursework

- Database Management
- Machine Learning
- Computer Networks
- Operating Systems
- Object-Oriented Programming
- Data Structures and Algorithms

Achievements

- **Algorithmic Proficiency:** Solved **100+** **LeetCode problems** focusing on optimized Data Structures and Algorithms. 
- **Technical Certifications:** Artificial Intelligence Foundation Certification • Machine Learning using Python (Infosys Springboard)
- Participated in the **72-hour national-level CodeSrijan Hackathon** at JUET Guna, building an **AI-powered resume screening and job matching system**. 